# AYAM BANJADE

Github Linkedin Portfolio

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Bremen, Germany


## PROFILE

Motivated and disciplined CS student passionate about technology and software development. As a fast learner, I am eager to tackle complex problems and adopt new tools and technologies.

## EDUCATION

**B.Sc. in Computer Science — minor in Mathematics**
(Sept. 2021 - Jun. 2024)

Constructor University Bremen

**Relevant Courses:**

- Software Engineering
- Databases and Web Services
- Algorithms and Data Structures
- Computer Networks
- Operating Systems
- Distributed Algorithms
- Secure and Dependable Systems
- Machine Learning
- Artificial Intelligence

## SKILLS

- C/C++, Rust, Java, Python, PHP, JavaScript, TypeScript, Flutter, Kotlin
- Svelte, jQuery, React, Next.js, HTML, CSS, Tailwind, npm
- MySQL, MongoDB, XML, JSON
- Linux, Shell Script, CLI, Git, Docker

## LANGUAGES

**English**: Native/Bilingual (C2)
**German**: Beginner (A2.2)

## EXPERIENCE

### Teaching Assistant at Constructor University Bremen

Numerical Methods — Feb. 2023 - Jun. 2023

- Created and solved challenging problems and helped 100+ students with numerical methods for root finding, interpolation, approximation, numerical differentiation / integration / differential equations, and their implementations in Python
- Streamlined grading process by developing Python scripts to transform and simplify complex CSV files

Programming in C/C++ — Sept. 2022 - Dec. 2022

- Provided comprehensive tutorials to 200+ students on programming concepts, including pointers, object-oriented programming, dynamic memory allocation, data structures, templates, macros, header files, and makefiles
- Debugged and validated C/C++ code, ensuring students' understanding and proficiency

Introduction to Computer Science — Sept. 2022 - Dec. 2022

- Coordinated and led tutorials addressing 95% of all CS students' inquiries in functional programming (Haskell), assembly programming, computer science foundations, discrete mathematics, Boolean algebra, Hoare logic, and digital circuit design

## LEADERSHIP ACTIVITIES

**Member of CS Hiring Commitee** — **Jun. 2023 - Present**
Constructor University Bremen

- Serving as the representative for the Computer Science student body in the hiring process for faculty, actively casting votes and advocating for the inclusion of student interests

**Member of Makers and Coders** — **Aug. 2019 - May 2021**
Ullens School (IBDP / High School)

- Coordinated seminars with STEM experts, conducted workshops for underprivileged Nepali students, and managed fundraising events for required equipment for schools and the club

## PROJECTS

### Rippit

- Developed a frontend and PHP backend of a forum website collaboratively using MySQL, MongoDB, jQuery, and VCS (git)
- Implemented post types, post rating, post sorting, user authentication, tickets (feedback, reports), and search suggestions, enhancing user engagement and content creation and discovery
- Utilized data definition languages to implement the entity-relationship model and query-dependent access rights using sanitized SQL queries for enhanced security and efficient data retrieval and analysis
- Designed homepage, individual post page, user profiles page, and admin users management

### Bug World

- Collaborated in agile sprints to build and improve different code bases for a full-stack simulation game in JavaScript and HTML/CSS
- Engineered and implemented a parser, assembler, simulator, and GUI using OOP principles
- Created automated test cases (Jest, Jasmine) to ensure robustness and functionality
- Authored and maintained detailed technical documentation and comprehensive specifications to ensure effective project communication and development

### Room Booking System

- Designed and built a Java-based, user-friendly, and intuitive CLI room booking system with file I/O, dynamic matching, and custom data structures for efficient data management

### Stochastic Methods Lab

- Developed stochastic models for financial mathematics using Python libraries (matplotlib, NumPy, pandas, SciPy) for CSV handling and complex dataset analysis and visualization

### Spam Email Extractor

- Built a C++ program to implement data extraction techniques to retrieve 57 relevant attributes from UCI's Spambase Data Set for effective spam email detection

Bremen, Germany, 2023-07-16 Ayam